Irregular Repetition Slotted ALOHA with Multi-Antenna Reception: Thresholds and Converse Bound

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***Abstract*—We study irregular repetition slotted ALOHA (IR-SA) with multi-antenna reception over Rayleigh block fading channels. An exact closed-form expression for the average decoding error probability $\bar{\epsilon}_{m,L}$ is derived for collision sizes $m = 1$ and $m = 2$ by applying the inclusion–exclusion principle, which generalizes known single-antenna results and remains valid for any finite number of antennas. Using this result, we develop a density-evolution-based analysis of multi-antenna IR-SA systems and characterize belief-propagation (BP) thresholds. Numerical results for the corresponding maximum a posteriori (MAP) decoding thresholds and converse bounds are also presented, demonstrating threshold saturation with spatial coupling.**

***Index Terms*—Belief propagation threshold, irregular repetition slotted ALOHA, multi-antenna reception, random access, Rayleigh fading, successive interference cancellation.**

I. Introduction

Irregular repetition slotted ALOHA (IR-SA) [1]–[7] is a powerful random access protocol that enables reliable communication in massive machine-type networks through packet repetition and successive interference cancellation (SIC). In this paper, we consider an IR-SA system with *multi-antenna reception*, where a large number of single-antenna devices communicates with a common access point (AP) equipped with L receive antennas over a slotted frame. The receiver performs iterative *spatio-temporal* SIC across antennas and slots, whereby the system performance is characterized in the asymptotic regime by the maximum normalized offered traffic that can be reliably supported.

For Rayleigh block fading channels with single-antenna reception ($L = 1$), the decoding process within a slot can be accurately characterized using threshold-based decoding with intra-slot SIC. Here m represents the number of packets colliding within a slot, and an exact closed-form expression for the average decoding error probability $\bar{\epsilon}_{m,1}$ in the presence of $m-1$ interferers is available [1], [2], which enables a rigorous density evolution (DE)-based analysis of IR-SA systems.

However, extending this analytical framework to multi-antenna reception ($L > 1$) is nontrivial, since the signal-to-interference-plus-noise ratio (SINR) statistics and decoding events become significantly more complex. As a result, existing analyses of multi-antenna IR-SA systems largely rely on simplified packet-erasure models [8]–[10] or numerical simulations [11], and an exact closed-form expression of the decoding error probability is missing. Note that Srivatsa *et al.* [12, Theorem 3] derived an expression for the average successful decoding probability $\theta_m = 1 - \bar{\epsilon}_{m,L\to\infty}$ under maximal-ratio-combining-based SIC (MRC-SIC). However, their result relies on an approximation and does not provide a closed-form expression for finite L.

In this paper, we derive an exact closed-form expression for the average decoding error probability $\bar{\epsilon}_{m,L}$ (for $m = 1$ and $m = 2$) for IR-SA systems with L-antenna reception over Rayleigh fading channels. Our analysis considers selection-combining-based SIC (SC-SIC), in which the receiver selects the strongest signal among the L antennas. By applying the inclusion-exclusion principle, we characterize the average decoding error probabilities under intra-slot SIC in the presence of antenna diversity. This result generalizes the known single-antenna expression $\bar{\epsilon}_{m,1}$ and enables a rigorous DE-based analysis of multi-antenna IR-SA systems, leading to the derivation of belief-propagation (BP) thresholds.

Once DE is characterized, the converse bound and the maximum a posteriori (MAP) decoding threshold can be derived via extrinsic information transfer (EXIT) chart area arguments [3], [13], [14]. Prior works on MAP thresholds and converse bounds have primarily focused on either collision-channel models [3], [13], [15] or single-antenna Rayleigh fading scenarios [2], [7]. In contrast, this paper addresses the MAP threshold and the converse bound for IR-SA systems with *multi-antenna reception over Rayleigh fading channels*.

II. System model

A. IR-SA Access Protocol with Multi-Antenna Reception

We consider an IR-SA system with multi-antenna reception, where N_{T} single-antenna devices communicate with a common AP equipped with L antennas over a frame of M time slots. Let $\mathcal{L} = \{1, 2, \ldots, L\}$ denote the set of antenna indices. The antennas at the AP are spaced sufficiently far apart to ensure independent fading values. In each frame, each device becomes active independently with probability $\pi \ll 1$ and, if active, generates one message that is encoded into a codeword (called a packet) using a common code $\mathcal{C}$ with code rate R. This encoder is implemented as a concatenation of

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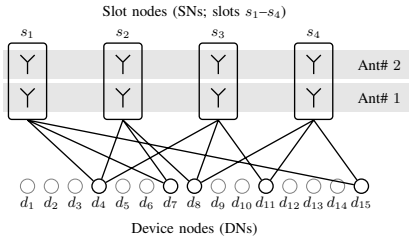


Fig. 1. Bipartite graph representation of multi-antenna IR-SA with $L = 2$, $M = 4$, $N_{\mathrm{T}} = 15$, $N_{\mathrm{a}} = 5$, and $\pi = N_{\mathrm{a}}/N_{\mathrm{T}} = 1/3$.

a channel encoder and a higher-order modulator. We restrict our analysis to $R \geq 1$; the case $R < 1$ is discussed in [16]. Each active device selects a repetition degree l with probability Λ_l according to the distribution $\{\Lambda_l\}_{l=1}^{l_{\max}}$. It then transmits l replicas of its packet in l randomly selected slots within a frame.

We define the normalized population size as $\alpha \triangleq N_{\mathrm{T}}/M$, which is constant. The average normalized offered traffic is $G = \pi N_{\mathrm{T}}/M = \pi\alpha$, representing the average number of packets transmitted per slot when the packet loss rate (PLR) is lower than a target value. We are interested in the asymptotic ($M \to \infty$) traffic G, which depends on probability distribution $\{\Lambda_l\}_{l=1}^{l_{\max}}$, given the average received signal-to-noise ratio (SNR), the code rate R and the number of antennas L.

B. Bipartite Graph and Degree Distributions

As illustrated in Fig. 1, the multi-antenna IR-SA protocol outlined in Sec. II-A can be characterized in a manner analogous to conventional single-antenna IR-SA [4], [5]. Specifically, it is modeled as a bipartite graph $\mathcal{G} = (\mathcal{D}, \mathcal{S}, \mathcal{E})$. Here, $\mathcal{D}$ denotes the set of N_{T} device nodes (DNs), $\mathcal{S}$ denotes the set of M slot nodes (SNs), and $\mathcal{E}$ represents the set of edges. An edge connects a DN $d_i \in \mathcal{D}$ to an SN $s_j \in \mathcal{S}$ if and only if the i-th device selects the j-th slot for transmission.

Importantly, even with multiple receive antennas, all multi-antenna processing, including intra- and inter-antenna SIC, is encapsulated within each slot node. As a result, the bipartite graph structure depends only on device transmission patterns and remains identical to that of the single-antenna case, consistent with prior work [12]. Consequently, multi-antenna effects enter the DE-based analysis solely through the slot-node decoding success probability $(1 - \bar{\epsilon}_{m,L})$ (see (15)).

The number of edges connected to a DN or SN is defined as the node degree. In this framework, a DN has degree l (i.e., it has l edges connected to SNs) with probability Λ_l. The set $\{\Lambda_l\}_{l=1}^{l_{\max}}$ is referred to as the DN degree distribution, where the average number of replicas per active device is given by

$$\bar{d} \triangleq \Lambda'(1) = \textstyle\sum_{l=1}^{l_{\max}} l\Lambda_l. \tag{1}$$

As $N_{\mathrm{T}} \to \infty$, the SN degree distribution is Poisson with $\Psi_n = \exp(-\alpha\bar{d})(\alpha\bar{d})^n/n!$ [4], [5].

The degree distributions are also defined from an edge-perspective in polynomials [5]: $\lambda(x) = \sum_{l=1}^{l_{\max}} \lambda_l x^{l-1}$ with $\lambda_l = \frac{l\Lambda_l}{\sum_{l=1}^{l_{\max}} l\Lambda_l}$ and

$$\rho(x) = \textstyle\sum_{n=1}^{N_{\mathrm{T}}} \rho_n x^{n-1} = \exp(-\alpha\bar{d}(1-x)), \tag{2}$$

with Poisson distribution

$$\rho_n = \frac{n\Psi_n}{\sum_{n=1}^{N_{\mathrm{T}}} n\Psi_n} = \exp(-\alpha\bar{d})\frac{(\alpha\bar{d})^{n-1}}{(n-1)!},\ n = 1, 2, \ldots. \tag{3}$$

C. Spatio-Temporal SIC Decoding in Rayleigh Block Fading

Decoding at the AP is an iterative decoding process, integrating both intra- and inter-slot as well as intra- and inter-antenna SICs.

1) Intra-Slot SIC: Assuming m packet replica collisions, the resultant L superimposed signals (one per antenna) within the slot are given by

$$\boldsymbol{y}_i = \textstyle\sum_{k\in\mathcal{K}} h_{k,i}\boldsymbol{x}_k + \boldsymbol{z}_i, \quad i = 1, 2, \ldots, L, \tag{4}$$

where $\mathcal{K}$ ($|\mathcal{K}| = m$) is the set of active devices. Here, device k is active and has accessed the slot. The vector $\boldsymbol{x}_k \in \mathcal{C}$ represents its codeword (or packet) of code $\mathcal{C}$. The vector $\boldsymbol{z}_i \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 I)$ is an additive circularly symmetric complex Gaussian (CSCG) noise with mean $\boldsymbol{0}$ and diagonal covariance matrix $\sigma^2 I$ with noise power σ^2 and identity matrix I.

The channel coefficient $h_{k,i}$ expresses the fading of the channel from the active device k to Ant# i of AP during the slot, which remains constant within a slot and varies independently across slots. For Rayleigh block fading channels, the received SNR $\gamma_{k,i} \triangleq P|h_{k,i}|^2/\sigma^2$ is exponentially distributed with $p_\gamma(x) = 1/\bar{\gamma} \cdot e^{-\frac{x}{\bar{\gamma}}},\ x > 0$, where $\bar{\gamma}$ is the average received SNR.

Decoding at AP within a slot involves an iterative process alternating between intra-antenna and inter-antenna SICs. It is assumed that the AP has full knowledge of the channel state information (CSI) and the number of packet collisions. We consider SC-SIC, in which SIC is performed independently at each antenna, and the antenna branch with the strongest received signal is selected for packet decoding.

a) Intra-Antenna SIC: Assuming an m-packet collision, at Ant# i, the SINR of the packet from device k, assuming $\gamma_{1,i} > \gamma_{2,i} > \cdots > \gamma_{m,i}$, can be calculated as

$$\eta_{k,i} \triangleq \frac{P|h_{k,i}|^2}{\sigma^2 + \sum_{u=k+1}^{m} P|h_{u,i}|^2} = \frac{\gamma_{k,i}}{1 + \sum_{u=k+1}^{m} \gamma_{u,i}}. \tag{5}$$

Assuming threshold-based decoding, the packet $\boldsymbol{x}_k$ is successfully decoded if the SINR $\eta_{k,i}$ exceeds the threshold η_0, i.e.,

$$\Pr\{\text{packet } \boldsymbol{x}_k \text{ decoded}\} = \begin{cases} 1, & \eta_{k,i} > \eta_0, \\ 0, & \eta_{k,i} \leq \eta_0, \end{cases} \tag{6}$$

where η_0 satisfies $R = \log_2(1 + \eta_0)$ for code $\mathcal{C}$ of rate R. Upon successful decoding, the packet is removed from the received signal $\boldsymbol{y}_i$. Subsequent decoding proceeds with the interference-removed signal. This iterative process is termed *intra-antenna* SIC.

Remark 1 (κ-MaxDecoding)**.** *Let κ denote the maximum number of colliding packets, either initial or residual, for which intra-antenna SIC is applied in a slot. Although intra-antenna SIC can be applied to collisions of arbitrary size due to the unbounded Poisson collision distribution, practical systems typically restrict SIC to low-order collisions because of interference, error propagation, and computational complexity*

[17]. Accordingly, we adopt the κ-MaxDecoding (e.g., $\kappa = 2$), in which intra-antenna SIC is performed only when the number of residual packets in a slot does not exceed κ. Hence, a slot originally containing an m-packet collision with $m > \kappa$ may still be decoded after inter-slot SIC if the residual degree is at most κ. □

An undecoded packet $\boldsymbol{x}_k$ with SNR above η_0, i.e., $\gamma_{k,i} > \eta_0$, is termed a *potentially decodable (PD)* packet [18], because it can be decoded upon removing the interfering packet $\boldsymbol{x}_{j\neq k}$ from signal $\boldsymbol{y}_i$.

b) Inter-Antenna SIC: The previously described intra-antenna SIC process enables the successful decoding of packets. These decoded packets are then canceled from the receive signals of other antennas within the same slot. This procedure is referred to as *inter-antenna* SIC.

The above two processes are repeated across all antenna elements until no further decoding is possible for any packet at any antenna element. This iterative decoding procedure is referred to as *intra-slot* SIC due to its execution within a slot.

2) Inter-Slot SIC: Following intra-slot SIC, once packets are successfully decoded, their replicas are removed from the specific slots identified by those packets. This processing, termed *inter-slot SIC*, facilitates subsequent intra-slot SIC on the remaining signals in those slots.

The intra- and inter-slot SIC processes can be modeled as the successive removal of edges in a bipartite graph $\mathcal{G}$. When a packet from a device (or a DN) is successfully decoded at at least one antenna in a slot (or a SN), the corresponding edge connected to the SN is removed via intra-slot SIC. Subsequently, all l edges incident to the same DN are removed through inter-slot SIC. This process iterates until no further edges can be eliminated.

III. Probabilities in Two-Collision (Single-Antenna Case)

In this section, we derive closed-form probability expressions for two packets being received in distinct SNR regions. These expressions will be used for computation of event probabilities in Sec. **??**.

A. Intra-Slot SIC

We consider a two-device fading channel with single-antenna reception, where the received signal at a slot is given by $\boldsymbol{y} = \sum_{k=1}^{2} h_k \boldsymbol{x}_k + \boldsymbol{z}$. In this context, the device k is active and has accessed the slot. .....

At this point, we summarize the possible events at a single-antenna reception for the packets sent from device l and device k, as shown in Table I.

Before deriving closed-form expressions for the probability of two packets being received in distinct SNR regions, we define $\Gamma = \{(\gamma_k, \gamma_l) | \gamma_k \in \Gamma_k, \gamma_l \in \Gamma_l\}$ as a probability region, with $\Gamma_k, \Gamma_l \subseteq \mathcal{R}^+$ being subsets of the non-negative real numbers $\mathcal{R}^+$.

For the two independent RVs γ_k and γ_l, the probability of being in region Γ is given by

$$P_\Gamma = \Pr\{(\gamma_k, \gamma_l) | \gamma_k \in \Gamma_k, \gamma_l \in \Gamma_l\} = \int_{\Gamma_l} p(\gamma_l) \int_{\Gamma_k} p(\gamma_k) \mathrm{d}\gamma_k \mathrm{d}\gamma_l.$$

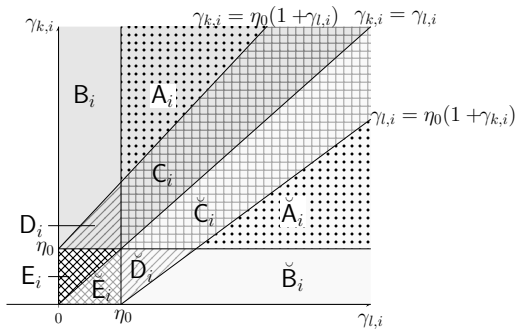


Fig. 2. The SIC feasible region in a two-device SA under $R \geq 1$.

The probability expressions are divided into two cases: $R \geq 1$ and $R < 1$, as detailed below.

B. Closed-Form Expressions of Probabilities ($R \geq 1$)

In the case of $R \geq 1$, Fig. 2 illustrates the practical regions A to E ($\gamma_k > \gamma_l$) along with their symmetric counterparts $\breve{\mathsf{A}}$ to $\breve{\mathsf{E}}$ ($\gamma_k < \gamma_l$). The events listed in Table I are defined.

For $\gamma_1 > \gamma_2$, the probabilities for regions A to E are:

$$P_{\mathsf{A}} = \Pr\{\mathsf{A} : (\gamma_1, \gamma_2) | \gamma_1 > \eta_0(1+\gamma_2), \gamma_2 > \eta_0\} \tag{7a}$$
$$P_{\mathsf{B}} = \Pr\{\mathsf{B} : (\gamma_1, \gamma_2) | \gamma_1 > \eta_0(1+\gamma_2), \gamma_2 \leq \eta_0\} \tag{7b}$$
$$P_{\mathsf{C}} = \Pr\{\mathsf{C} : (\gamma_1, \gamma_2) | \gamma_2 < \gamma_1 \leq \eta_0(1+\gamma_2), \gamma_2 > \eta_0\} \tag{7c}$$
$$P_{\mathsf{D}} = \Pr\{\mathsf{D} : (\gamma_1, \gamma_2) | \eta_0 < \gamma_1 \leq \eta_0(1+\gamma_2), \gamma_2 \leq \eta_0\} \tag{7d}$$
$$P_{\mathsf{E}} = \Pr\{\mathsf{E} : (\gamma_1, \gamma_2) | \gamma_2 < \gamma_1 \leq \eta_0, \gamma_2 \leq \eta_0\}. \tag{7e}$$

For $\gamma_1 < \gamma_2$, the probabilities for regions $\breve{\mathsf{A}}$ to $\breve{\mathsf{E}}$ are defined analogously. In Appendix I, we derive the closed-form expressions for probabilities P_{A} through P_{E} while excluding $P_{\breve{\mathsf{A}}}$ and $P_{\breve{\mathsf{E}}}$ due to symmetry.

C. Closed-Form Expressions of Probabilities ($0 < R < 1$)

In the case of $R < 1$, the decoding threshold η_0 is less than 1, i.e., $\eta_0 < 1$. This implies that the two lines $\gamma_1 = \gamma_2$ and $\gamma_1 = \eta_0(1+\gamma_2)$ (or $\gamma_2 = \eta_0(1+\gamma_1)$) intersect at a point denoted by $(\frac{\eta_0}{1-\eta_0}, \frac{\eta_0}{1-\eta_0})$ (see Fig. **??**), where $\frac{\eta_0}{1-\eta_0}$ is a unique solution to equation $\gamma_1 = \eta_0(1+\gamma_1)$ (or $\gamma_2 = \eta_0(1+\gamma_2)$).

This implies that the probability that both packets are decodable over the entire γ_1–γ_2 plane is given by

$$\Pr\left\{\mathsf{A}' \cup \breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'\right\} = (P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\breve{\mathsf{F}}'}) + (P_{\breve{\mathsf{A}}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\mathsf{F}'}).$$

Note that the ranges of the double integral used to compute $P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'}$ are the same as those in (7a).

In order to integrate both cases of R, as in the case of $R \geq 1$, we give the probabilities for the case $\gamma_1 > \gamma_2$:

$$\begin{aligned} P_{(\mathsf{A}' \cup \mathsf{F}')} &= P_{\mathsf{A}' \cup \mathsf{F}' \cup \breve{\mathsf{F}}'} - P_{\breve{\mathsf{F}}'} \\ &= \Pr\{(\gamma_1, \gamma_2) | \gamma_1 > \eta_0(1+\gamma_2), \gamma_2 > \eta_0\} \\ &\quad - \Pr\left\{(\gamma_1, \gamma_2) | \eta_0(1+\gamma_2) < \gamma_1 < \gamma_2, \gamma_2 > \frac{\eta_0}{1-\eta_0}\right\}, \\ P_{\mathsf{C}'} &= \Pr\left\{(\gamma_1, \gamma_2) | \gamma_2 < \gamma_1 < \eta_0(1+\gamma_2), \eta_0 < \gamma_2 < \frac{\eta_0}{1-\eta_0}\right\}. \end{aligned}$$

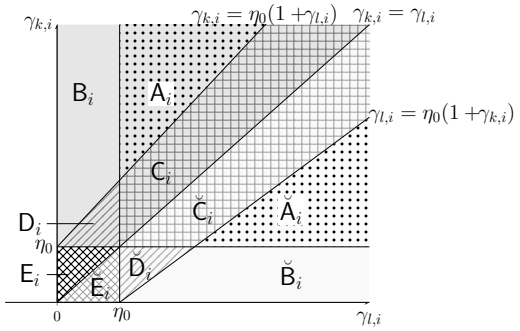


Fig. 3. The SIC feasible region in a two-collision within a slot ($R \geq 1$).

TABLE I
EVENTS AT A SINGLE ANTENNA, WHERE P_k AND P_l ARE PACKETS SENT BY DEVICE k AND l, RESPECTIVELY.

region	event
A ($\breve{\mathsf{A}}$)	P_k (P_l) is decoded first and then P_l (P_k) is decoded.
B ($\breve{\mathsf{B}}$)	P_k (P_l) is decoded, while P_l (P_k) is undecoded and PD.
C, $\breve{\mathsf{C}}$	P_k and P_l are PD.
D ($\breve{\mathsf{D}}$)	P_k (P_l) is undecoded and PD; P_l (P_k) is undecoded and not PD.
E, $\breve{\mathsf{E}}$	P_k and P_l are undecoded and not PD.

The probabilities for region ($\breve{\mathsf{A}}'\cup\breve{\mathsf{F}}'$) and $\breve{\mathsf{C}}'$ ($\gamma_1 < \gamma_2$) can be derived in a similar manner, and are omitted for brevity. We derive the closed-form expressions of $P_{(\mathsf{A}'\cup\mathsf{F}')}$ and $P_{\mathsf{C}'}$ in [19]. Note that the closed-form expressions of P_{B}, P_{D}, and P_{E} remain invariant with respect to R.

The received SNRs from devices k and l at Ant# i are denoted by $\gamma_{k,i}$ and $\gamma_{l,i}$, respectively. The random variables (RVs) $\gamma_{k,i}$, $\gamma_{l,i}$, $\gamma_{k,j}$, and $\gamma_{l,j}$ ($k,l \in \mathcal{D}$, $k \neq l$ and $i,j \in \mathcal{L}$, $i \neq j$) are mutually independent. The integral regions illustrated in Fig. 3 are introduced to clarify the computation. For example, the probability associated with the region A_i is denoted as P_{A_i}. Due to symmetry, we have $P_{\mathsf{\Gamma}_i} = P_{\breve{\mathsf{\Gamma}}_i} \triangleq P_{\mathsf{\Gamma}}$, $\mathsf{\Gamma} \in \{\mathsf{A},\mathsf{B},\mathsf{C},\mathsf{D},\mathsf{E}\}$. The extended version of this paper [16] provides the closed-form expressions for these probabilities, which serve as the basis for the calculations in the L-antenna case.

IV. AVERAGE ERROR PROBABILITY WITHIN A SLOT

Consider a generic slot t with m packets received by L antennas, as in (4), and select one packet as the reference. We derive closed-form expressions for $\bar{\epsilon}_{m,L}$ for $m = 1, 2$ under intra-slot SIC performed across and within antennas. These results form the analytical basis for the DE framework developed in the next section.

In threshold-based decoding, a single decoding step is defined as the recovery of a packet at any of the L antennas, followed by its cancellation from all antennas. Consequently, for an m-collision, up to m successive steps may be required. Let $D(m,r,L)$ denote the probability that the reference packet is successfully decoded specifically at step r. The average decoding error probability for the reference packet in the presence of $m-1$ interferers within a slot using L antennas is

$$\bar{\epsilon}_{m,L} = 1 - \textstyle\sum_{r=1}^{m} D(m,r,L). \tag{8}$$

In deriving the results for the L-antenna case, we rely on the following fundamental facts:

(i) RVs $\gamma_{k,i}$ are independent and identically distributed.

(ii) The inclusion-exclusion principle for computing the probability of the union of L sets $A_1,\ldots,A_L$ [20]:

$$\Pr(\cup_{\ell=1}^{L} A_\ell) = \sum_{\emptyset\neq\mathcal{S}\subseteq\{1,\ldots,L\}} (-1)^{|\mathcal{S}|+1} \Pr(\cap_{\ell\in\mathcal{S}} A_\ell). \tag{9}$$

(iii) The binomial theorem: $\sum_{k=0}^{n} \binom{n}{k} x^{n-k}y^k = (x+y)^n$.

A. Closed-form Expression for a Single-Packet Slot ($m = 1$)

The probability that the packet of device k, which is the only packet received in the slot, is successfully decoded by at least one of the antennas, is given by

$$D(1,1,L) = 1 - \prod_{i=1}^{L}(1 - \Pr\{\gamma_{k,i} > \eta_0\}) = 1 - (1 - e^{-\frac{\eta_0}{\bar{\gamma}}})^L.$$

B. Closed-form Expression for a Two-Collision Slot ($m = 2$)

1) $D(2,1,L)$: We derive the probability that the reference packet is successfully decoded in the first threshold-based decoding step. This probability is expressed as

$$D(2,1,L) = \frac{1}{2}\Pr\{\Upsilon\} = \frac{1}{2}\Big(1 - \{1 - 2(P_{\mathsf{A}} + P_{\mathsf{B}})\}^L\Big). \tag{10}$$

Here, $\Pr\{\Upsilon\}$ denotes the probability that either P_k or P_l is decodable at any of the L antennas in the first step. Note that the reference packet is either P_k or P_l with probability $1/2$. In the following, we deduce the two probabilities and thus (10).

For simplicity, we first consider the case where the reference packet P_k (or P_l) is decoded at a specific antenna $i \in \mathcal{L}$ before P_l (or P_k). The probability of this event is given by $\Pr\{\Upsilon_i\} = \Pr\{\mathsf{A}_i\} + \Pr\{\mathsf{B}_i\} + \Pr\{\breve{\mathsf{A}}_i\} + \Pr\{\breve{\mathsf{B}}_i\} = 2(P_{\mathsf{A}} + P_{\mathsf{B}})$. Consequently, the probability that the reference packet is first decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Upsilon\} = \Pr\{\bigcup_{i\in\mathcal{L}} \Upsilon_i\}$. With the inclusion-exclusion principle (9) and the binomial theorem, we obtain

$$\begin{aligned}
\Pr\{\Upsilon\} = \Pr\{\bigcup_{i\in\mathcal{L}} \Upsilon_i\} &= \sum_{\emptyset\neq\mathcal{S}\subseteq\mathcal{L}} (-1)^{|\mathcal{S}|+1} \Pr\left(\bigcap_{i\in\mathcal{S}} \Upsilon_i\right) \\
&= \sum_{\ell=1}^{L} \sum_{\substack{\mathcal{S}\subseteq\mathcal{L} \\ |\mathcal{S}|=\ell}} (-1)^{\ell+1} \Pr\left(\bigcap_{i\in\mathcal{S}} \Upsilon_i\right) \\
&= \sum_{\ell=1}^{L} (-1)^{\ell+1} \binom{L}{\ell} \{2(P_{\mathsf{A}}+P_{\mathsf{B}})\}^\ell \\
&= 1 - \{1 - 2(P_{\mathsf{A}} + P_{\mathsf{B}})\}^L.
\end{aligned} \tag{11}$$

Here, the expression is transformed by noting that, for a fixed $\ell > 0$, there are $\binom{L}{\ell}$ subsets $\mathcal{S} \subseteq \mathcal{L}$ such that $|\mathcal{S}| = \ell$, and by using the assumption that the SNRs at the antennas are i.i.d.

Using $\Pr\{\Upsilon\}$, we obtain $D(2,1,L)$ as given in (10).

2) $D(2,2,L)$: Next, we derive the probability that the reference packet is successfully decoded in the second threshold-based decoding step. This probability is expressed as

$$\begin{aligned} D(2,2,L) =& \frac{1}{2}\big(\Pr\{\Delta\}+\Pr\{\Xi\}\big) \\ =& \frac{1}{2}\Big\{1+2(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L \\ & \text{–}2(1\text{–}2P_{\mathsf{A}}\text{–}P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L\text{–}(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}})^L\Big\}. \end{aligned} \tag{12}$$

Here, $\Pr\{\Delta\}$ denotes the probability that two packets are decoded at the same antenna in $\mathcal{L}$. In contrast, $\Pr\{\Xi\}$ represents the probability that the packets are successfully decoded across at least two distinct antennas. This probability is derived by considering all possible antenna combinations for two scenarios: 1) P_l and P_k are decoded independently at distinct antennas, respectively; and 2) the initial decoding of P_l (P_k) in an antenna set $\mathcal{L}_1 \subset \mathcal{L}$ with $|\mathcal{L}_1| = \ell_1$ and its subsequent cancellation via inter-antenna SIC enables the decoding of P_k (P_l) in a distinct set $\mathcal{L}_2 \subset \mathcal{L}\setminus\mathcal{L}_1$ with $|\mathcal{L}_2| = \ell_2$. Moreover, the reference packet decoded in the second step is either P_k or P_l with probability $1/2$.

In the following, we derive these two probabilities. We first consider the case where both packets are decoded at a single antenna $i \in \mathcal{L}$. The probability of this event is given by $\Pr\{\Delta_i\} = \Pr\{\mathsf{A}_i\} + \Pr\{\breve{\mathsf{A}}_i\} = 2P_{\mathsf{A}}$. Consequently, the probability that P_k and P_l are successfully decoded at any antenna in $\mathcal{L}$ is $\Pr\{\Delta\} = \Pr\{\bigcup_{i\in\mathcal{L}}\Delta_i\}$. Using the inclusion-exclusion principle (9) and the binomial theorem, we obtain

$$\begin{aligned} \Pr\{\Delta\} = \Pr\{\bigcup_{i\in\mathcal{L}}\Delta_i\} =& \sum_{\ell=1}^{L}(-1)^{\ell+1}\binom{L}{\ell}(2P_{\mathsf{A}})^{\ell} \\ =& 1-(1-2P_{\mathsf{A}})^L. \end{aligned}$$

The derivation is analogous to (11) and is omitted for brevity.

Next, we derive the probability $\Pr\{\Xi\}$ that the two packets are successfully decoded at a different antenna. To this end, consider the event $\Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}$, which represents the situation where P_l is independently decodable only at Ant# i, P_k is decodable or in a PD state at Ant# j, and neither packet is decodable at Ant# p.

This configuration implies that (1) P_l and P_k are independently and successfully decoded at Ants# i and # j, respectively, and (2) the initial decoding of P_l at Ant# i, followed by its subsequent removal at Ant# j through inter-antenna SIC, enables the exclusive decoding of P_k.

For $i \neq j \neq p$, this event can be expressed as follows:

$$\Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k} = \breve{\mathsf{B}}_i \cap \{\mathsf{B}_j \cup \mathsf{C}_j \cup \breve{\mathsf{C}}_j \cup \mathsf{D}_j\} \cap \{\overline{\mathsf{A}_p \cup \breve{\mathsf{A}}_p}\}. \tag{13}$$

Define $\mathcal{L}_1 \subset \mathcal{L}$ as the subset of antenna elements where only P_l can be independently decodable. Define $\mathcal{L}_2 \subseteq \mathcal{L}\setminus\mathcal{L}_1$ as the subset where P_k is decodable or in PD state within each antenna element. Further, let $\mathcal{L}_3 = \mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)$ as the subset comprising the remaining antenna elements that cannot decode either P_l or P_k. Let $\ell_n = |\mathcal{L}_n|$, $n = 1,2,3$. Note that $\mathcal{L}_1\cup\mathcal{L}_2\cup\mathcal{L}_3 = \mathcal{L}$ and $\mathcal{L}_{n_1}\cap\mathcal{L}_{n_2} = \emptyset$ for $n_1 \neq n_2$.

For any $i \in \mathcal{L}_1$, $j \in \mathcal{L}_2$, and $p \in \mathcal{L}_3$, the intersection of sets (13) is written as

$$\begin{aligned} \bigcap_{\substack{i\in\mathcal{L}_1, j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k} =& \bigcap_{i\in\mathcal{L}_1}\breve{\mathsf{B}}_i \bigcap_{j\in\mathcal{L}_2}\{\mathsf{B}_j\cup\mathsf{C}_j\cup\breve{\mathsf{C}}_j\cup\mathsf{D}_j\} \bigcap_{p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}\{\overline{\mathsf{A}_p\cup\breve{\mathsf{A}}_p}\} \\ =& P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2} \\ \triangleq& \Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}. \end{aligned}$$

Using the inclusion-exclusion principle (9), we have

$$\begin{aligned} \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\} =& \Pr\{\bigcup_{\substack{i,j,p\in\mathcal{L},\\ i\neq j\neq p}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\} \\ =& \sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ \mathcal{L}_1\neq\emptyset}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\setminus\mathcal{L}_1\\ \mathcal{L}_2\neq\emptyset}}(-1)^{|\mathcal{L}_1|+|\mathcal{L}_2|+2}\Pr\{\bigcap_{\substack{i\in\mathcal{L}_1,j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\} \\ =& \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\sum_{\substack{\mathcal{L}_1\subset\mathcal{L}\\ |\mathcal{L}_1|=\ell_1}}\sum_{\substack{\mathcal{L}_2\subseteq\mathcal{L}\setminus\mathcal{L}_1\\ |\mathcal{L}_2|=\ell_2}}\Pr\{\bigcap_{\substack{i\in\mathcal{L}_1,j\in\mathcal{L}_2,\\ p\in\mathcal{L}\setminus(\mathcal{L}_1\cup\mathcal{L}_2)}} \Xi_{ijp}^{\mathsf{P}_l\to\mathsf{P}_k}\} \\ =& \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\}. \end{aligned}$$

The term $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}$ accounts for the number of distinct antenna pairs $(\mathcal{L}_1,\mathcal{L}_2)$ of sizes ℓ_1 and ℓ_2, respectively, such that $\mathcal{L}_1$ and $\mathcal{L}_2$ are disjoint subsets of $\mathcal{L}$.

The symmetrical property of the SNRs for P_k and P_l ensures that $\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}$. The probability that P_k and P_l are independently and successfully decoded at each antenna element in their respective $\mathcal{L}_1$ and $\mathcal{L}_2$ is $\Pr\{\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi_{\mathcal{L}_1\mathcal{L}_2}^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} = P_{\mathsf{B}}^{\ell_1+\ell_2}(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}$.

Therefore, we have

$$\begin{aligned} \Pr\{\Xi\} =& \Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\}+\Pr\{\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\}-\Pr\{\Xi^{\mathsf{P}_l\to\mathsf{P}_k}\cap\Xi^{\mathsf{P}_l\leftarrow\mathsf{P}_k}\} \\ =& \sum_{\ell_1=1}^{L-1}\sum_{\ell_2=1}^{L-\ell_1}(-1)^{\ell_1+\ell_2}\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2}\xi_{\Xi}(\ell_1,\ell_2), \end{aligned}$$

where

$$\xi_{\Xi}(\ell_1,\ell_2) = (2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_2} - P_{\mathsf{B}}^{\ell_1+\ell_2})(1-2P_{\mathsf{A}})^{L-\ell_1-\ell_2}.$$

Define $\ell_3 = \ell_1+\ell_2$ with $\ell_3 \in \{2,3,\ldots,L\}$. It follows that $\binom{L}{\ell_1}\binom{L-\ell_1}{\ell_2} = \binom{L}{\ell_3}\binom{\ell_3}{\ell_1}$. Thus, $\Pr\{\Xi\}$ can be rewritten as

$$\begin{aligned} \Pr\{\Xi\} =& \sum_{\ell_1=1}^{L-1}\sum_{\ell_3=\ell_1+1}^{L}(-1)^{\ell_3}\binom{L}{\ell_3}\binom{\ell_3}{\ell_1}\xi_{\Xi}(\ell_1,\ell_3-\ell_1) \\ =& \sum_{\ell_3=2}^{L}\sum_{\ell_1=1}^{\ell_3-1}(-1)^{\ell_3}\binom{L}{\ell_3}\binom{\ell_3}{\ell_1} \\ & \cdot(2P_{\mathsf{B}}^{\ell_1}(P_{\mathsf{B}}+2P_{\mathsf{C}}+P_{\mathsf{D}})^{\ell_3-\ell_1}-P_{\mathsf{B}}^{\ell_3})(1-2P_{\mathsf{A}})^{L-\ell_3}. \end{aligned}$$

Finally, by applying the binomial theorem, we obtain

$$\begin{aligned} \Pr\{\Xi\} =& 2(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L - 2(1\text{–}2P_{\mathsf{A}}\text{–}P_{\mathsf{B}}\text{–}2P_{\mathsf{C}}\text{–}P_{\mathsf{D}})^L \\ & \text{–}(1\text{–}2P_{\mathsf{A}}\text{–}2P_{\mathsf{B}})^L + (1\text{–}2P_{\mathsf{A}})^L. \end{aligned}$$

With $\Pr\{\Delta\}$ and $\Pr\{\Xi\}$ we obtain $D(2,2,L)$ in (12).

Remark 2. *Focusing on the case $m = 2$, the average decoding error probability $\bar{\epsilon}_{m,L}$ for MRC-SIC in [12, Theorem 3] is*

TABLE II
AVERAGE DECODING ERROR PROBABILITY $\bar{\epsilon}_{m,L}$ $(m = 1, 2)$ WITHIN A SLOT. GAP DENOTES THE ABSOLUTE DIFFERENCE $|\text{Sim.} - \text{Theo.}|$. SIMULATION RESULTS REPORTED AS $< 10^{-6}$ INDICATE THAT NO DECODING ERROR WAS OBSERVED OVER 10^6 MONTE-CARLO TRIALS. $(\eta_0 = 10, \bar{\gamma} = 10\,\text{dB})$.

		MRC-SIC [12, Theorem 3]			SC-SIC (8)		
m	L	Sim.	Theo.	Gap	Sim.	Theo.	Gap
1	4	1.88×10^{-2}	1.90×10^{-2}	1.7×10^{-4}	1.60×10^{-1}	1.60×10^{-1}	1.0×10^{-4}
	8	1.0×10^{-5}	1.0×10^{-5}	$< 10^{-6}$	2.53×10^{-2}	2.55×10^{-2}	2.0×10^{-4}
	16	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	7.0×10^{-4}	7.2×10^{-4}	1.8×10^{-5}
	32	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	4.2×10^{-7}	4.2×10^{-7}
	64	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$	1.8×10^{-13}	1.8×10^{-13}
2	4	6.16×10^{-1}	7.32×10^{-1}	1.2×10^{-1}	7.89×10^{-1}	7.89×10^{-1}	5.1×10^{-5}
	8	3.68×10^{-1}	4.81×10^{-1}	1.1×10^{-1}	5.84×10^{-1}	5.84×10^{-1}	9.7×10^{-5}
	16	1.41×10^{-1}	2.07×10^{-1}	6.6×10^{-2}	3.31×10^{-1}	3.31×10^{-1}	1.7×10^{-4}
	32	2.25×10^{-2}	3.84×10^{-2}	1.6×10^{-2}	1.10×10^{-1}	1.09×10^{-1}	3.9×10^{-4}
	64	7.0×10^{-4}	1.3×10^{-3}	6.0×10^{-4}	1.19×10^{-2}	1.19×10^{-2}	3.1×10^{-5}

obtained via an approximation that is accurate only in the large-L regime. In contrast, the proposed SC-SIC scheme provides an exact closed-form expression for $\bar{\epsilon}_{m,L}$ that is valid for any finite L. As shown in Table II, the SC-SIC analytical results closely match the simulation results for all considered finite L, while the MRC-SIC approximation exhibits a noticeable mismatch. □

V. DENSITY EVOLUTION WITH L-ANTENNA RECEPTION

In this section, building on [1], [2], [4] and the closed-form results of the previous section, we derive two DE iteration equations that characterize the asymptotic performance $(M, N_{\mathrm{T}} \to \infty)$ of multi-antenna IR-SA systems.

We refer to the DNs (SNs) as the DN (SN) group. In the protograph, let ${}^{\ell}p$ be the probability that an edge incident on the DN group carries an erasure message towards SNs for the ℓ-th iteration. Similarly, let ${}^{\ell}q$ be the probability that an edge incident on the SN group carries an erasure message towards the DN groups.

First, consider the DN group, which has degree distribution $\lambda(x)$. In the ℓ-th iteration, an edge incident to a degree-l DN is removed if at least one of the other $l-1$ edges has been successfully resolved. Given the device activation probability π, the average probability that a DN-to-SN message remains erased is expressed as [2]–[4]:

$${}^{\ell}p = \pi \textstyle\sum_{l=1}^{l_{\max}} \lambda_l \cdot {}^{(\ell-1)}q^{l-1} = \pi\lambda({}^{(\ell-1)}q). \tag{14}$$

Second, consider the SN group, which has degree distribution $\rho(x)$. The probability that an edge carries an erasure message from the SN group to the DN group in the ℓ-th iteration is ${}^{\ell}q = \sum_{r=1}^{\infty} \rho_r \cdot {}^{\ell}q^{(r)}$. Here ρ_r is the fraction of edges connected to an SN of degree-r (see (3)), and ${}^{\ell}q^{(r)}$ is the probability that an edge carries an erasure message, given that it is connected to the SN of degree-r. The erasure probability of the degree-r SN is [2]

$${}^{\ell}q^{(r)} = 1 - \textstyle\sum_{m=1}^{r} \binom{r-1}{m-1} (1-{}^{\ell}p)^{r-m} ({}^{\ell}p)^{m-1} (1-\bar{\epsilon}_{m,L}). \tag{15}$$

Here, $(1 - \bar{\epsilon}_{m,L})$ denotes the probability that the replica associated with an outgoing edge is successfully decoded when the reduced degree of the SN is m. In other words, in a slot where r replicas collide, $r - m$ replicas are removed via inter-slot SIC, after which the receiver attempts to decode a randomly selected replica among the remaining m replicas using L receive antennas. The closed-form expression of $\bar{\epsilon}_{m,L}$ is provided in Sec. IV.

With some manipulations, we have [1], [2]

$${}^{\ell}q = 1 - e^{-\alpha\bar{d}\,{}^{\ell}p} \textstyle\sum_{m=0}^{\infty} \frac{(\alpha\bar{d}\,{}^{\ell}p)^m}{m!} (1 - \bar{\epsilon}_{m+1,L}). \tag{16}$$

Substituting for (14) with $G = \pi\alpha$, we have, with ${}^{(\ell=0)}q = 1$,

$${}^{\ell}q = 1 - e^{-G\bar{d}\lambda({}^{(\ell-1)}q)} \sum_{m=0}^{\infty} \frac{(G\bar{d}\lambda({}^{(\ell-1)}q))^m}{m!} (1 - \bar{\epsilon}_{m+1,L}). \tag{17}$$

Let ${}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0) = \lim_{\ell} {}^{(\ell)}q$ be the convergence value of the iteration. The PLR of the systems is [1], [4]

$$\begin{aligned}\mathrm{PLR}(G, \bar{d}, L, \bar{\gamma}, \eta_0) &= \Lambda({}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0)) \\ &= \textstyle\sum_{l=1}^{l_{\max}} \Lambda_l \left({}^{(\infty)}q(G, \bar{d}, L, \bar{\gamma}, \eta_0)\right)^l. \end{aligned} \tag{18}$$

Given target PLR^*, the asymptotic iterative decoding threshold, also called the BP decoding threshold, is defined as the supremum offered traffic value [4] [1]:

$$G^{\mathrm{BP,blk}} = \sup\{G : \mathrm{PLR}(G, \bar{d}, L, \bar{\gamma}, \eta_0) \to 0\}. \tag{19}$$

Under the κ-MaxDecoding in Remark 1, the summation in (17) is restricted to residual degrees $m \le \kappa - 1$ (e.g., $\kappa = 2$).

VI. CONVERSE AND MAP THRESHOLD

A. Converse Bound

We are now ready to describe an asymptotic threshold upper bound on average offered traffic G^{C} of graph-based IR-SA systems with multi-antenna reception. The upper bound is referred to as a converse bound in information theory representing a theoretical limit to be approached.

By omitting the iterative index for simplification, we rewrite the iteration equations (14) and (**??**) of the density evolution:

$$\begin{aligned} p &= \pi\lambda(q) \triangleq f_{\mathrm{d}}(q), && q \in (0, 1] \\ q &= 1 - \chi(1-p) \triangleq f_{\mathrm{s}}(p), && p \in (0, \pi]. \end{aligned} \tag{20}$$

Note that the initial value of p is activation probability π. An example of an EXIT chart is shown in Fig. **??**.

Function $f_{\mathrm{d}}(q)$ $(f_{\mathrm{s}}(p))$ is also called the average EXIT function of DNs (SNs) [21] [22]. Let the areas below the DN (SN) EXIT functions over interval $[0, 1]$ $([0, \pi])$ be

$$A_{\mathrm{d}} = \int_0^1 f_{\mathrm{d}}(q)\mathrm{d}q \text{ and } A_{\mathrm{s}} = \int_0^{\pi} f_{\mathrm{s}}(p)\mathrm{d}p. \tag{21}$$

A necessary condition for successful decoding is an open tunnel between the two curves in the EXIT chart [21] [14], which displays $f_{\mathrm{d}}(q)$ and $f_{\mathrm{s}}^{-1}(q)$. Such area matching means that (Fig. **??**)

$$A_{\mathrm{d}} + A_{\mathrm{s}} < \pi. \tag{22}$$

The area below the DN EXIT function is [2]

$$A_{\mathrm{d}} = \pi \sum_l \lambda_l \int_0^1 q^{l-1}\mathrm{d}q = \pi \sum_l \frac{\lambda_l}{l} = \frac{\pi}{\Lambda'(1)} = \frac{\pi}{\bar{d}}. \tag{23}$$

On the other hand, the area below the SN EXIT function is

$$A_{\mathrm{s}} = \pi - \int_0^{\pi} \chi(1-z)\mathrm{d}z = \pi - \tau(\pi). \tag{24}$$

Under the 2-MaxDecoding constraint, $\tau(x)$ is expressed as [2]

$$\begin{aligned}\tau(x) &\triangleq \int_0^x \chi(1-z)\mathrm{d}z \\ &= \frac{1}{\alpha\bar{d}}\sum_{m=0}^{1}\left((1-\bar{\epsilon}_{m+1,L})(1-e^{-\alpha\bar{d}x}\sum_{k=0}^{m}\frac{(\alpha\bar{d}x)^k}{k!})\right)\end{aligned} \tag{25}$$

is expressed in a closed form. In (f) we used Gaussian integral [23, Eq. (2.321.1)]:

$$\int x^n e^{ax}\mathrm{d}x = e^{ax}\left(\sum_{k=0}^{n}(-1)^k\frac{n!}{a^{k+1}(n-k)!}x^{n-k}\right)+C.$$

Combining (23) and (24), we have $\pi\alpha < \alpha\bar{d}\tau(\pi)$.

With $G=\pi\alpha$, the solution of equation $\pi\alpha=\alpha\bar{d}\tau(\pi)$, i.e.,

$$G^{\mathrm{C,blk}}=\sum_{m=0}^{1}\left((1-\bar{\epsilon}_{m+1,L})(1-e^{-G^{\mathrm{C,blk}}\bar{d}}\sum_{k=0}^{m}\frac{(G^{\mathrm{C,blk}}\bar{d})^k}{k!})\right) \tag{26}$$

gives the converse bound of asymptotic threshold $G^{\mathrm{C,blk}}$, which is a function of the average number of transmissions $\bar{d}$ and is independent of the specific DN degree distribution.

B. MAP Threshold

We derive threshold $G^{\mathrm{MAP,blk}}$ under MAP decoding for graph-based multi-antenna IR-SA systems over Rayleigh block fading channels. MAP decoding resembles that of LDPC code over BEC where activation probability π is analogous to the channel erasure probability in the LDPC context [3].

By omitting the iterative index for simplification, we rewrite the iteration equations (14) and (**??**) of the density evolution:

$$\begin{aligned} p &= \pi\lambda(q), && q\in(0,1] \\ q &= 1-\chi(1-p), && p\in(0,\pi]. \end{aligned} \tag{27}$$

Note that the initial value of p is activation probability π. Erasure probability p emitted by the DNs tends to a limit value, called x, which is a fixed point of equations

$$x \;=\; \pi\lambda(1-\chi(1-x)). \tag{28}$$

Solving this fixed point equation for $\pi\in(0,1]$, we get

$$\pi(x)=\frac{x}{\lambda(1-\chi(1-x))},\quad x\in(0,\bar{x}], \tag{29}$$

where $\pi(\bar{x})=1$ and $0<\bar{x}<1$. In the LDPC context [21, Theorem 3.59], $\pi(x=1)=1$, since every check node always has two more edges. Thus the constant term in the polynomial of the node-perspective distribution of the check nodes is $\rho_{l=1}=0$. It follows that $\rho(0)=0$. In IR-SA systems, however, $\rho_{l=1}\neq 0$ (3), and thus $\rho(0)\neq 0$, and $\chi(0)\neq 0$ (**??**).

At the fixed point, if the erasure probability emitted from the DN is x, then the PLR of the systems equals $\Lambda(1-\chi(1-x))$. Similar to MAP decoding in the LDPC context [21, Lemma 3.116], we define BP EXIT function $h^{\mathrm{BP}}(\pi)$ in parametric form by the following curve :

$$\begin{aligned}&(\pi(x),h^{\mathrm{BP}}(\pi(x))) \\ &=\begin{cases}(\pi,0), & \pi\in[0,\pi^{\mathrm{BP}}), \\ (\pi(x),\Lambda(1-\chi(1-x))), & \pi\in(\pi^{\mathrm{BP}},1]\leftrightarrow x\in(x^{\mathrm{BP}},\bar{x}].\end{cases}\end{aligned}$$

The BP EXIT curve is a trace of this parameter equation for x starting at $x=\bar{x}$ until $x=x^{\mathrm{BP}}$. Indeed, function $\pi(x)$ has a unique minimum, which determines threshold $\pi^{\mathrm{BP}}=\pi(x^{\mathrm{BP}})$ [21, Lemma 3.116] [24].

The integral under the curve $(\pi(x),\Lambda(1-\chi(1-x)))$ is called a trial entropy associated with (λ,χ) [21, Definition 3.119]:

$$\begin{aligned}P(x)&=\int_0^x h^{\mathrm{BP}}(\pi(z))\mathrm{d}\pi(z)=\int_0^x\Lambda(1-\chi(1-z))\mathrm{d}\pi(z)\\ &=\Lambda(1-\chi(1-x))\pi(x)-\Lambda'(1)\int_0^x z\chi'(1-z)\mathrm{d}z\\ &=\frac{x\Lambda(1-\chi(1-x))}{\lambda(1-\chi(1-x))}+\Lambda'(1)\left(x\chi(1-x)-\tau(x)\right),\end{aligned} \tag{30}$$

where we use (1), (29), and $\tau(x)$ is defined as

$$\begin{aligned}\tau(x) &\triangleq \int_0^x \chi(1-z)\mathrm{d}z \\ &= \frac{1}{\alpha\bar{d}}\sum_{m=0}^{1}\left((1-\bar{\epsilon}_{m+1,L})(1-e^{-\alpha\bar{d}x}\sum_{k=0}^{m}\frac{(\alpha\bar{d}x)^k}{k!})\right).\end{aligned}$$

Note that MAP threshold π^{MAP} is the solution of $P(x)=0$ and defines $\pi^{\mathrm{MAP}}=\pi(x^{\mathrm{MAP}})$, and the BP and MAP EXIT curves coincide above π^{MAP} [21, Theorem 3.120]. Therefore, given $\lambda(x)$ and $\chi(x)$, we find the MAP threshold by computing

$$\int_{\pi^{\mathrm{MAP}}}^{\pi(\bar{x})} h^{\mathrm{BP}}(\pi)\mathrm{d}\pi = P(\bar{x})-P(x^{\mathrm{MAP}})=P(\bar{x}). \tag{31}$$

As a result, the MAP threshold of graph-based multi-antenna IR-SA systems can be upper bounded:

$$G^{\mathrm{MAP,blk}}=\alpha\pi^{\mathrm{MAP}}. \tag{32}$$

VII. Numerical and Simulation Results

In this section, using the closed-form expressions of $\bar{\epsilon}_{m,L}$ and DE under the ($\kappa=2$)-MaxDecoding assumption, we present numerical results for the BP thresholds in (19), along with the corresponding MAP thresholds and converse bounds. The derivations of the MAP thresholds and converse bounds are omitted due to space limitations and can be found in [16].

Figure 4 compares the PLR performance of multi-antenna IR-SA using the MRC-SIC and our proposed SC-SIC (18) for $L=8$ at a target $\mathrm{PLR}^*=10^{-2}$. For reference, the performance of the single-antenna IR-SA ($L=1$) [1] is also included. The same degree distribution, optimized specifically for the proposed multi-antenna IR-SA via a differential evolution algorithm [25], is applied to all schemes. Simulation results with a finite number of slots ($M=10^5$) for SC-SIC closely match the asymptotic DE analysis, thereby validating our analytical framework; the small residual gap is attributable to the finite number of slots in the simulation. In contrast, the MRC-SIC approach in [12] exhibits a noticeable discrepancy between simulation and analysis. This is because its analysis is approximate for $m\geq 2$, as indicated by the results in Table II.

Figure 6 plots the BP thresholds $G^{\mathrm{BP,blk}}$ versus the average received SNR $\bar{\gamma}$ for $L=8$ and $L=16$. In the low-SNR regime ($\bar{\gamma}<12$ dB), MRC-SIC achieves higher BP thresholds due to its combining gain, which is essential for

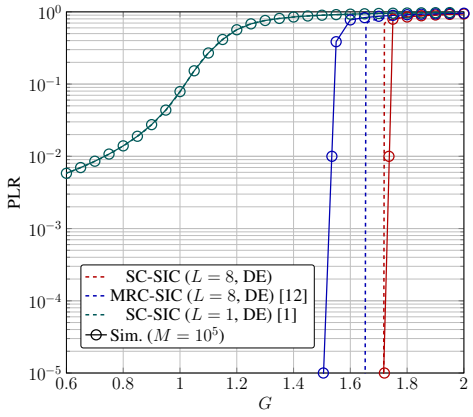


Fig. 4. PLRs vs. offered traffic G of IR-SA for the MRC-SIC [12] and the proposed SC-SIC: $\eta_0 = 5$, $\bar{\gamma} = 20$ dB, and $\Lambda(x) = 0.586x^2 + 0.291x^3 + 0.031x^6 + 0.022x^7 + 0.006x^8 + 0.063x^{10}$.

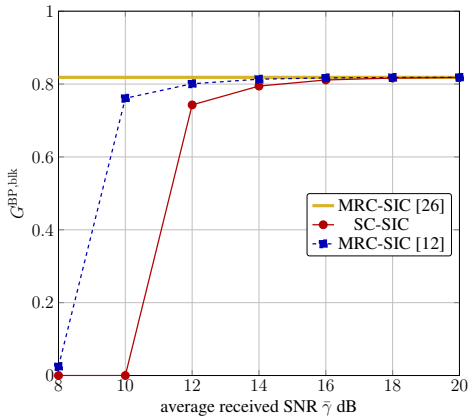


Fig. 5. ($\kappa = 1$):BP thresholds $G^{\mathrm{BP,blk}}$ vs. average received SNR $\bar{\gamma}$ for MRC-SIC [12] and SC-SIC; $\Lambda(x) = x^3$, $\eta_0 = 10$, and $L = 3$.

raising the effective SNR above the decoding threshold η_0. By contrast, the lack of combining gain in SC-SIC causes the SINRs across all L antennas to remain below the decoding threshold, thereby leading to a complete decoding failure. In the high-SNR regime ($\bar{\gamma} \geq 12$ dB), a performance crossover occurs; while intra-antenna SIC likely succeeds on at least one antenna even without combining, MRC-SIC indiscriminately aggregates interference from all antennas, thereby limiting the effective SINR.

Once the DE iterative equations are obtained, the MAP threshold and the converse bound for the L-antenna IR-SA can be evaluated [16]. Table III summarizes these thresholds and bounds. In addition, the L-antenna spatially coupled scheme [16], extended from the single-antenna case [2], [3], yields the BP thresholds $G^{\dagger\,\mathrm{BP,Conv}}$, demonstrating threshold saturation.

VIII. Conclusion

This paper developed an analytical framework for IR-SA with multi-antenna reception over Rayleigh block fading channels. Exact closed-form expressions for the average decoding error probability $\bar{\epsilon}_{m,L}$ under SC-SIC enabled a rigorous DE-based analysis without altering the bipartite graph structure.

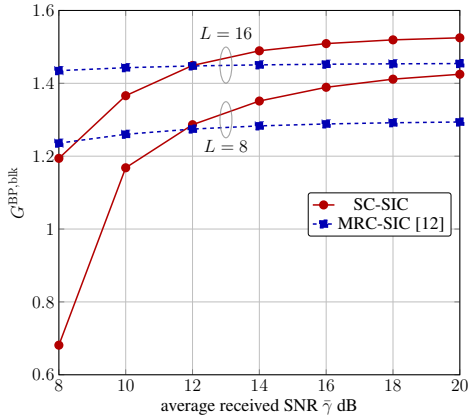


Fig. 6. ($\kappa = 2$):BP thresholds $G^{\mathrm{BP,blk}}$ vs. average received SNR $\bar{\gamma}$ for MRC-SIC [12] and SC-SIC; $\Lambda(x) = x^3$, $\eta_0 = 10$, and $L = 8, 16$.

TABLE III
BP thresholds of the spatially coupled scheme, $G^{\dagger\,\mathrm{BP,Conv}}$, MAP thresholds of d-regular repetition SA, $G^{\mathrm{MAP,blk}}_{\mathrm{reg}}$, and IR-SA, $G^{\mathrm{MAP,blk}}$, as well as the converse bound $G^{\mathrm{C,blk}}$ ($\eta_0 = 1$, $\bar{\gamma} = 20$ dB).

L	$\bar{d}$, (d)	$G^{\dagger\,\mathrm{BP,Conv}}$	$G^{\mathrm{MAP,blk}}_{\mathrm{reg}}$	$G^{\mathrm{MAP,blk}}$	$G^{\mathrm{C,blk}}$
1 [2]	3.103 (3)	1.824	1.824	1.826	1.829
2	3.066 (3)	1.969	1.969	1.969	1.972
3	3.068 (3)	1.976	1.976	1.976	1.979

The obtained BP and MAP thresholds, along with the converse bounds, demonstrated threshold saturation via spatial coupling and elucidated the performance trade-offs between SC-SIC and MRC-SIC. Future work will address higher-order collisions, alternative combining schemes, and more general fading models.

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APPENDIX I
CLOSED-FORM PROBABILITY EXPRESSIONS

A. $R \geq 1$

We present analytical probability expressions for the regions shown in Fig. 3. Due to symmetry, we focus on the range where $\gamma_k > \gamma_l$.

We derive the probabilities P_{A} to P_{E} as follows.

$$P_{\mathsf{A}} = \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0}, \quad P_{\mathsf{B}} = \frac{e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{C}} = \frac{1}{2}e^{-\frac{2\eta_0}{\bar{\gamma}}} - \frac{e^{-\frac{1}{\bar{\gamma}}((1+\eta_0)^2-1)}}{1+\eta_0},$$

$$P_{\mathsf{D}} = \frac{\eta_0 e^{-\frac{\eta_0}{\bar{\gamma}}}}{1+\eta_0} - e^{-\frac{2\eta_0}{\bar{\gamma}}} + \frac{e^{-\frac{\eta_0(2+\eta_0)}{\bar{\gamma}}}}{1+\eta_0}, \; P_{\mathsf{E}} = \frac{1}{2}\left(1 - e^{-\frac{\eta_0}{\bar{\gamma}}}\right)^2.$$

B. $0 < R < 1$

We present analytical expressions of $P_{(\mathsf{A}'\cup\mathsf{F}')}$ and $P_{\mathsf{C}'}$, whose corresponding regions are shown in Fig. **??**.

For region $\breve{\mathsf{F}}'$, We have the probability:

In order to integrate both cases of R, as in the case of $R \geq 1$, we give the probabilities for region $(\mathsf{A}'\cup\mathsf{F}')$ and C' ($\gamma_1 > \gamma_2$):

$$\begin{aligned}
P_{(\mathsf{A}'\cup\mathsf{F}')} =& P_{\mathsf{A}'\cup\mathsf{F}'\cup\breve{\mathsf{F}}'} - P_{\breve{\mathsf{F}}'} \\
=& \Pr\left\{\mathsf{A}'\cup\mathsf{F}'\cup\breve{\mathsf{F}}' : (\gamma_1,\gamma_2) | \gamma_1 > \eta_0(1+\gamma_2), \gamma_2 > \eta_0\right\} \\
& - \Pr\left\{\breve{\mathsf{F}}' : (\gamma_1,\gamma_2) | \eta_0(1+\gamma_2) < \gamma_1 < \gamma_2, \gamma_2 > \frac{\eta_0}{1-\eta_0}\right\} \\
=& \frac{e^{-\frac{2\eta_0}{\bar{\gamma}(1-\eta_0)}}}{(1-\eta_0)}\left(e^{\frac{{\eta_0}^2(1+\eta_0)}{\bar{\gamma}(1-\eta_0)}} - \frac{1-\eta_0}{2}\right), \qquad (33a) \\
P_{\mathsf{C}'} =& \Pr\left\{\mathsf{C}' : (\gamma_1,\gamma_2) | \gamma_2 < \gamma_1 < \eta_0(1+\gamma_2), \eta_0 < \gamma_2 < \frac{\eta_0}{1-\eta_0}\right\} \\
=& \frac{e^{-\frac{2\eta_0}{\bar{\gamma}_0(1-\eta_0)}}(1-\eta_0)}{2(1+\eta_0)} - e^{-\frac{2\eta_0}{\bar{\gamma}}}\left(\frac{e^{-\frac{{\eta_0}^2}{\bar{\gamma}}}}{1+\eta_0} - \frac{1}{2}\right). \qquad (33b)
\end{aligned}$$

Here, the probabilities for region $(\breve{\mathsf{A}}' \cup \breve{\mathsf{F}}')$ and $\breve{\mathsf{C}}'$ ($\gamma_1 < \gamma_2$) are omitted due to symmetry.

We derive the closed-form expressions of the probabilities $P_{(\mathsf{A}'\cup\mathsf{F}')}$ and $P_{\mathsf{C}'}$. Note that the closed-form expressions of P_{B}, P_{D}, and P_{E} are invariant with respect to R.

Finally, given that $P_{(\mathsf{A}'\cup\mathsf{F}')}$ in (33a) and P_{A} in (7a) both represent the probability of two packets being successfully decoded, we use P_{A} for simplicity, extending it to the case $0 < R < 1$. Similarly, we replace $P_{\mathsf{C}'}$ in (33b) with P_{C} for $0 < R < 1$.